



23ST-BCAP-MA-05-12001

Seat No.

01540

BACHELOR OF COMPUTER APPLICATION Examination
BCA SEMESTER-5 NOVEMBER 2025 (NEP) NOVEMBER - 2025

CS -30: PROGRAMMING IN PYTHON

Faculty Code : 003

Subject Code : 23SI-BCAP-MA-05-12001

Time : 2 Hours]

[Total Marks : 50

- Q.1 Attempt the following. 3
- (A) (1) What is the output of 50/20 and 50//20?
(2) What is the output of type('python')?
(3) Python commands are evaluated/executed in _____.
- Q.1 Attempt any one of the following. 2
- (B) (1) Explain tuple with example.
(2) Explain list with example.
- Q.1 Attempt any one of the following. 5
- (C) (1) Explain Dictionary in python with example.
(2) Explain function and scoping with suitable example.
- Q.2 Attempt the following. 3
- (A) (1) Define: ADT in python.
(2) _____ statement forces exception to occur.
(3) Partition and exchange sort is _____.
- Q.2 Attempt any one of the following. 2
- (B) (1) Explain self in python with example.
(2) Explain assertion with example.
- Q.2 Attempt any one of the following. 5
- (C) (1) Explain Selection sort with example. ✓
(2) Explain Bubble sort with example. ✓
- Q.3 Attempt the following. 3
- (A) (1) Dynamic programming is mainly an optimization over plain _____
(2) You are given a knapsack that can carry a maximum weight of 60. There are 4 items with weights {20, 30, 40, 70} and values {100, 80, 90, 200}. What is the maximum value of the items you can carry using the knapsack?
(3) _____ function is used to save a plot as a PNG file in PyLab.

- Q.3 Attempt any one of the following. 2
- (B) (1) Explain legend() and show() function.
(2) Explain Memoization in python.
- Q.3 Attempt any one of the following. 5
- (C) (1) Explain Plotting Mortgages with example.
(2) Explain divide and conquer in detail.
- Q.4 Attempt the following. 3
- (A) (1) TCL stands for _____.
(2) The _____ protocol is used to send email over the internet.
(3) Define: IP address.
- Q.4 Attempt any one of the following. 2
- (B) (1) What is an entry field?
(2) Explain widget attributes.
- Q.4 Attempt any one of the following. 5
- (C) (1) Explain UDP Server and UDP Client with example.
(2) Write a python program to download a web page from Internet.
- Q.5 Attempt the following. 3
- (A) (1) List out connection() function parameters.
(2) The _____ method of the cursor class is used to fetch only one row from the result set of a query.
(3) Define: MySQL database.
- Q.5 Attempt any one of the following. 2
- (B) (1) Write a python code to Retrieving All Rows from a Table.
(1) Write a python code to Updating Rows in a Table.
- Q.5 Attempt any one of the following. 5
- (C) (1) Write a python code to creating Python database and book table like book_id, book_name, author, year and price through python.
(2) Write a python code to read book name from the user and update that record otherwise give message "book name is not available, try with another book name".